

TableTask Movesense QC Questions

Comparison of full-session QC, first-2-minute QC, and RR odd-even reliability

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1. Purpose and source files

This document answers two QC questions raised after the latest meeting with Veronica:

- Question 1: Are the problematic full-session files and first-2-minute files the same participants/sessions?
- Question 2: Are poor/problematic files tied to a specific physical sensor, or can any sensor be poor or good depending on the session?

The answer separates three levels that should not be mixed: dyad/session level, sensor-recording level, and physical sensor-ID level.

Important note: Step 2 full-session QC labels are visual review labels documented in the workflow guide. Step 2b labels are first-2-minute QC flags from the Step 2b output. Step 3a reliability labels are numeric RR odd-even Pearson reliability labels. The thresholds in Step 3a are provisional review labels, not final exclusion thresholds.

Source file	Use in this note
TableTask_ECG_HR_Workflow_Guide_through_Step3_v4_clean_organized.docx	Source for Step 2 full-session visual QC lists and dyad-level labels.
3a_pretrial_2min_sensor_rr_reliability.csv	Source for Step 2b first-2-minute clean/caution/poor flags and pretrial RR reliability.
3a_sensor_id_reliability_summary.csv	Source for physical sensor-ID reliability summaries across full session, pretrial, and trial windows.
3a_session_reliability_summary.csv	Source for session-level RR reliability summaries.
3a_rr_reliability_run_summary.txt	Source for Step 3a overall reliability counts.

2. Question 1: Are the same files problematic in the full session and in the first 2 minutes?

Direct answer: At the dyad/session level, yes. The same 9 dyads are poor/problematic in both Step 2 full-session visual QC and Step 2b first-2-minute QC. At the sensor-recording level, the overlap is almost exact: all 12 sensors that are poor in the first 2 minutes are also poor in the full-session QC, but there are 2 additional sensors that are poor in the full-session QC and only caution in the first 2 minutes.

Level	Step 2 full-session poor	Step 2b first-2-minute poor	Overlap	Full-session poor only	First-2-minute poor only
Dyad/session	9 dyads	9 dyads	9 dyads	0 dyads	0 dyads
Sensor recording	14 sensors	12 sensors	12 sensors	2 sensors	0 sensors

Interpretation: The poor dyads were already problematic during the first two minutes, before task movement. This supports Veronica's concern that a substantial portion of the noise is not only caused by table carrying. It likely also reflects sensor contact, placement, strap tightness, electrode wetting, or recording setup.

Recording folder	Dyad ID	Step 2 full-session QC	Step 2b first-2-minute QC	Poor in both?
2026_03_03-19_31_30	P03_P04	Poor/problematic	Poor	Yes
2026_03_03-22_32_43	P05_P06	Poor/problematic	Poor	Yes
2026_03_03-23_45_18	P07_P08	Poor/problematic	Poor	Yes
2026_03_26-18_28_36	P17_P18	Poor/problematic	Poor	Yes
2026_03_27-18_34_14	P23_P24	Poor/problematic	Poor	Yes
2026_03_31-18_39_54	P25_P26	Poor/problematic	Poor	Yes
2026_03_31-21_32_11	P27_P28	Poor/problematic	Poor	Yes
2026_04_07-17_29_26	P33_P34	Poor/problematic	Poor	Yes
2026_04_07-22_33_07	P39_P40	Poor/problematic	Poor	Yes

Sensor-recording difference: The two records below were poor/problematic in the full-session visual QC but were only caution during the first two minutes. This suggests later task-related artifact or signal degradation across the recording, rather than an already poor first-2-minute baseline.

Recording folder	Sensor ID	Role	Step 2 full-session QC	Step 2b first-2-minute QC	Interpretation
2026_04_07-17_29_26	242930002204	A	Poor/problematic	Caution	First two minutes were not fully poor, but the full session became problematic.
2026_04_07-22_33_07	242930002204	B	Poor/problematic	Caution	First two minutes were not fully poor, but the full session became problematic.

3. Question 2: Are poor files linked to one physical sensor?

Direct answer: No, poor files are not explained by one physical sensor only. Some sensor IDs are more concerning than others, but the two most frequently reused sensors also produced many clean/reliable recordings. This means the problem is likely a combination of sensor/device variability and session-level setup factors.

Visual QC counts by physical sensor ID are summarized below. Values are shown as clean/caution/poor.

Sensor ID	Step 2 uses	Step 2 full-session visual QC: clean/caution/poor	Step 2b uses	Step 2b first-2-minute QC: clean/caution/poor
242930002204	14	6/4/4	14	10/2/2
242930002205	2	0/1/1	2	1/0/1
242930002208	14	7/2/5	14	8/1/5
251430002238	1	1/0/0	1	1/0/0
251430002239	2	0/0/2	2	0/0/2
251430002247	1	0/0/1	1	0/0/1
251430002277	2	0/1/1	2	1/0/1

Numeric RR reliability by physical sensor ID is summarized below. Values are shown as n: reliable/caution/low. The Step 3a thresholds are provisional: reliable $r \geq 0.75$, caution $0.50 \leq r < 0.75$, and low $r < 0.50$. The trial-window row is most relevant for future Step 4 decisions because Step 4 analyzes dyad-by-trial windows.

Sensor ID	Full-session reliability	Pretrial reliability	Trial-window reliability	Trial mean r	Trial min r
242930002204	14: 9/3/2	14: 7/3/4	126: 102/6/18	0.794	-0.189
242930002205	2: 0/1/1	2: 1/0/1	17: 8/0/9	0.434	-0.126
242930002208	14: 8/3/3	14: 7/3/4	126: 83/13/30	0.726	-0.323
251430002238	1: 1/0/0	1: 0/0/1	9: 4/3/2	0.670	0.126
251430002239	2: 0/0/2	2: 0/0/2	13: 4/0/9	0.472	0.112
251430002247	1: 0/0/1	1: 0/0/1	9: 0/2/7	0.164	-0.207
251430002277	2: 2/0/0	2: 1/1/0	12: 11/0/1	0.870	0.411
Sensor ID	Interpretation				
242930002204	Frequently reused. It can produce clean and poor recordings; most trial windows are reliable, so this does not look like a uniformly defective sensor.				
242930002205	Only two recordings. Mixed/concerning: no full-session reliable label and 9/17 trial windows are low reliability.				
242930002208	Frequently reused. It can produce clean data, but it has more low trial-window reliability than 242930002204 and should be monitored.				
251430002238	Only one recording. Full-session reliable but pretrial low; not enough uses to infer a device-level pattern.				
251430002239	Consistently concerning across its two uses: full-session and pretrial reliability are low in both, and most trial windows are low.				
251430002247	Only one recording, and it is poor/low. This is concerning for that recording but insufficient to conclude a device problem.				
251430002277	Two uses. Reliability index is mostly strong, although visual QC flagged one session; this suggests sensor ID alone is not enough for exclusion.				

4. Overall conclusion and recommended wording

Overall conclusion: The poor/problematic dyads are the same in the full-session and first-2-minute QC summaries. At the sensor-recording level, Step 2b poor sensors are a subset of full-session poor sensors. The first-2-minute QC therefore captures most of the poor recordings and supports the interpretation that many problems were already present before the task movement began.

Sensor conclusion: The problem is not limited to a single physical sensor. Sensor 251430002239 is consistently concerning across its two uses. Sensor 251430002247 is also poor/low but appears only once, so it cannot establish a repeated device-level pattern. Sensor 242930002205 is mixed but concerning. The two most common sensors, 242930002204 and 242930002208, both produce clean and poor recordings; however, 242930002208 has more low trial-window reliability than 242930002204. This suggests that setup/contact/movement factors matter, and sensor identity alone should not determine exclusion.

Recommended wording for Veronica:

I compared the full-session QC with the first-2-minute QC. At the dyad level, the same 9 dyads were poor in both analyses. At the sensor-recording level, all 12 sensors that were poor in the first two minutes were also poor in the full-session QC. Two additional sensors were poor in the full-session QC but only caution during the first two minutes, suggesting possible later task-related artifact or signal degradation. I also checked whether the poor files were tied to specific physical sensor IDs. The issue is not explained by only one sensor: frequently reused sensors can be clean or poor depending on the session. However, sensor 251430002239 is consistently concerning across its two uses, and 242930002208 appears somewhat less stable than 242930002204 by trial-window RR reliability. This supports checking sensor placement, strap tightness, and electrode wetting before task onset, while also continuing to track physical sensor ID in the QC file.

5. Reliability-index context

The new Step 3a reliability output supports the same interpretation. The script successfully loaded Step 2, Step 2b, Step 3, and analysis-unit files and computed:

Reliability level	Rows computed	Reliable	Caution	Low
Full-session sensor reliability	36	20	7	9
First-2-minute pretrial sensor reliability	36	16	7	13
Trial-window sensor reliability	312	212	24	76
Trial-window dyad reliability	156	86	12	58
Main-analysis eligible trial-window dyads	115	64	8	43

Interpretation: This index is single-sensor RR reliability, not dyadic synchrony. For dyad-window QC, the relevant value is the minimum of the two sensors because dyadic analysis is only as trustworthy as the weaker signal. These reliability labels should be reviewed with Veronica before they are used as final exclusion thresholds.